

Model	Baseline	New Feature(s) Added	MAE	RMS E	R ²	Notes
Linear Regression	Original features	+ age, area, bathrooms	45,200	55,000	0.52	Significant improvement
Random Forest	Original features	+ age, area, bathrooms	16,900	27,200	0.885	Slight improvement
Gradient Boosting	Original features	+ age, area, bathrooms	17,300	27,800	0.880	Minor improvement
XGBoost	Original features	+ age, area, bathrooms	18,200	30,500	0.855	Moderate improvement

New Features Added:

- House age = YrSold - YearBuilt
- Total area = GrLivArea + TotalBsmtSF
- Total bathrooms = FullBath + 0.5 × HalfBath

Notes

Linear Regression

Linear Regression showed the most improvement because it relies heavily on feature quality and cannot automatically capture complex relationships.

Random Forest / Gradient Boosting

Tree-based models showed only slight improvements since they can already capture nonlinear patterns in the data.

XGBoost

XGBoost improved moderately, suggesting that additional features still provided useful information.